

Duong Phuc Ta

Ph.D. Candidate, Department of Agricultural and Applied Economics, Virginia Tech

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FIELDS & RESEARCH FOCUS

Primary: Applied Econometrics; Agricultural Economics **Secondary:** Environmental and Resource Economics

I study how climate shocks, water scarcity, infrastructure, and policy incentives affect land-use decisions, agricultural outcomes, and environmental adaptation, using causal inference — difference-in-differences, instrumental variables, and double machine learning — combined with household survey, panel, and spatial data.

EDUCATION

- **Virginia Tech** 2023 – Expected May 2028
Ph.D. in Agricultural and Applied Economics USA
- **University of Greenwich** 2014 - 2015
MSc in Business and Financial Economics United Kingdom
- **The National Economics University** 2004 - 2008
Bachelor in Economics Vietnam

JOB MARKET PAPER

“Estimating impacts of climate stress on US milk production: a panel data with debiased machine learning approach”

Duong Phuc Ta (first author). *Current job market paper.*

This paper uses panel data and debiased machine learning to estimate the impacts of climate stress on U.S. milk production.

PUBLICATIONS

Journal Articles in English:

- [1] Bradford Mills, Chi Ta, **Duong Phuc Ta**, Huong Nguyen (2025). “Salinity Inundation, Profitability, and Rice Farming Exits in Mekong Delta of Vietnam”. *Journal of the Agricultural and Applied Economics Association*, 4(2), pp:170-186. DOI: [10.1002/jaa2.70004](https://doi.org/10.1002/jaa2.70004)
- [2] Bradford Mills, Dung Phuong Le, **Duong Phuc Ta**, Lien Nhu, Danh Thanh Vo and Ricardo Labarta (2022). “Intensive and extensive rice farm adaptations in salinity-prone areas of the Mekong Delta”. *Climate and Development*, 15(2), pp:162-176. DOI: [10.1080/17565529.2022.2072800](https://doi.org/10.1080/17565529.2022.2072800)
- [3] Nguyen Dinh Chuc and **Ta Phuc Duong** (2019). “Trade, Structural Adjustments and Productivity Growth in Vietnam: The Shift to Services”. *Journal of Southeast Asian Economies*, 36(2), pp:256-273.
- [4] **Ta Phuc Duong** (2015). “Effects of Capital Flows on Housing Prices in Emerging Economies: A Panel Approach”. *Vietnam’s Socio-Economic Development*, No. 81 (April 2015).

Journal Articles (in Vietnamese) — published under the name Ta Phuc Duong:

- [1] **Ta Phuc Duong** (2023). “Welfare impacts of non-farm livelihood participation of households in Vietnamese rural area”. *Journal of Economic Studies*, 12(523).
- [2] **Ta Phuc Duong** (2021). “Impacts of agricultural portfolio changes on farm household income under the context of climate changes in Vietnam Mekong Delta”. *Journal of Economic Studies*, 12(523).
- [3] **Ta Phuc Duong** and Nguyen Thi Anh Tuyet (2018). “Trends in economic functions of Vietnamese households”. *Journal of Economic Studies*, 7(482).
- [4] **Ta Phuc Duong** (2017). “Regional and industrial specialization of labour in Vietnam”. *Journal of Regional Sustainable Development*, 12.

WORKING PAPERS & RESEARCH IN PROGRESS

- [1] **Duong Phuc Ta** (first author). “Estimating impacts of climate stress on US milk production: a panel data with debiased machine learning approach”. *Current job market paper*.
- [2] **Duong Phuc Ta**, Bradford Mills, Kurt Stephenson, John Fike. “Economic Feasibility and Carbon Sequestration of Silvopasture in the Southeastern U.S.” *Submitted to a journal and under review*.

CONFERENCE PRESENTATIONS

- [1] **Duong Phuc Ta**, Bradford Mills, Kurt Stephenson, John Fike, Tom Stanley (2025). “Economic Feasibility and Carbon Sequestration of Silvopasture in the Southeastern U.S.” *Presented at the AAEA-WAEA Annual Meeting*, July 27-29, 2025, Denver, CO.
- [2] Nguyen Chien Thang and **Ta Phuc Duong** (2019). “Changing trend of production function of Vietnamese households”. RC06-VSA International Conference: The Family in Modern and Global Societies: Persistence and Change. October 17-19, 2019, Hanoi, Vietnam.

WORK EXPERIENCE

- **Virginia Tech** [🌐] 2023 – present
Graduate Research Assistant USA
- **The International Center for Tropical Agriculture (CIAT), Asia Hub** [🌐] 2018 – 2022
Research Assistant / Data Analyst (held concurrently with the VIE research post) Vietnam
 - Contributed to salinity-intrusion impact assessment in the Mekong Delta (2019–2022) and cassava production and value-chain research in Vietnam (2019–2021).
 - Developed questionnaires and CSPro survey programs, supervised data collection, cleaned and analyzed data, and contributed to co-authored research.
- **Vietnam Institute of Economics (VIE)** [🌐] 2010 – 2023
Researcher Vietnam
 - Led projects on the dynamics of farm and non-farm employment in Vietnam’s rural economy (2022) and agricultural structure under climate change in the Mekong Delta (2021), from proposal development through analysis and final reports.
 - Contributed to research on household economic functions, labour productivity, structural transformation, agricultural value chains, and farm-firm linkages.

TEACHING, TRAINING & MENTORING EXPERIENCE

- **Virginia Tech** [🌐] 2023 – present
Graduate Teaching Assistant USA

PROFESSIONAL SERVICE & AFFILIATIONS

Affiliations: Agricultural and Applied Economics Association (AAEA).

SKILLS

- **Statistical & Econometric Software:** Stata, R, Python, Matlab, SQL, LaTeX
- **Geospatial & Remote Sensing:** Google Earth Engine
- **Machine Learning & Data Science:** Scikit-learn, TensorFlow, Keras
- **Reproducibility & Collaboration:** Git, GitHub, Quarto, VSCode
- **Research Methods:** Econometrics, Causal Inference, Microeconomic Methods, Cost-Benefit Analysis, Machine Learning, Survey Design, Experimental Design, Data Management, Data Visualization
- **Additional Training:** Machine Learning Specialization, Stanford University via Coursera (3 courses), 2025

LANGUAGES

Vietnamese (native), English (fluent).

REFERENCES

1. **Bradford Mills**
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Department of Agricultural and Applied Economics, Virginia Tech
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2. **Wei Zhang**
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